Electric Charges & Fields

1. A polythene piece, rubbed with wool, is found to have negative charge of 4×10^{-7} the number of electrons transferred from wool to polythene is
   1) 1.5×10^{12}
   2) 2.5×10^{12}
   3) 2.5×10^{13}
   4) 3.5×10^{13}
2. If a body has positive charge on it, then it means it has
   1) Gained some protons
   2) Lost some protons
   3) Gained some electrons
   4) Lost some electrons
3. What will be the total charge stored in the system, if +q, +2q, +3q, +4q,....(up to +20q)
   1) +20q
   2) +420b
   3) +210q
   4) +190q
4. Which of the following is not the property of charge?
   1) Charge is additive
   2) A charge is self-destructive
   3) Quantization of charge
   4) Charge is conserved
5. Which of the following is not true about electric charge?
   1) Chare on a body is always integral multiple of certain charge known as charge of electrons
   2) Charge is a scalar quantity
   3) Net charge on an isolated system is always conserved
   4) Charge can be converted into energy and energy can be converted into charge
6. A total charge Q is broken in two parts Q_1 and Q_2 and they are placed at a distance R form each other. The maximum force of repulsion between them will occur, when:
   1) $Q_2=\frac{Q}{R}, Q_1=Q-\frac{Q}{R}$
   2) $Q_2=\frac{Q}{4}, Q_1=Q-\frac{2Q}{3}$
   3) $Q_2=\frac{Q}{R}, Q_1=\frac{3Q}{4}$
   4) $Q_2=\frac{Q}{R}, Q_2=\frac{Q}{2}$
7. Two charges +2 C and +6C are repelling each other with a force of 12 N. If each charge is given -2 C of charge, then the value of the force will be:
   1) 4N (Attractive)

2) 4N (Repulsive)
3) 8N (Repulsive)
4) Zero

8. Two charges $2\mu C$ and $8\mu C$ are separated by 6 cm. The neutral point is at:

1) 4 cm from 2μC
2) 2 cm from 2μC
3) 2 cm from 8μC
4) 3 cm from 8μC

9. Two point charges A and B, having charges +Q and -Q respectively, are placed at certain distance apart and the force acting between them is F. If 25% charge of A is transferred to B, them the force between the charges becomes:

1) $\frac{4F}{3}$
2) F
3) $\frac{9F}{16}$
4) $\frac{16F}{9}$

10. The unit of permittivity of free space ε_0 is.

1) Newton metre²/Coulomb²
2) Coulomb²/ Newton metre²
3) Coulomb²/(Newton metre)²
4) Coulomb/Newton metre

11. Electric charges of 1μC, - 1μC and 2μC are placed in air at the corners A, B and C respectively of an equilateral triangle ABC having length of each side 10 cm. The resultant force on the charge at C is.

1) 0.9 N
2) 1.8 N
3) 27 N
4) 3.6 N

12. Point charges +4q, -q and +4q are kept on the x-axis at points x = 0,x = a and x = 2a respectively, then.

1) Only-q is in stable equilibrium
2) None of the charges are in equilibrium
3) All the charges are in unstable equilibrium
4) All the charges are in the stable equilibrium

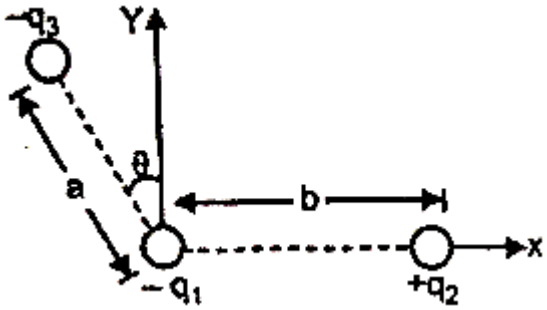
13. Four charges equal to -Q are placed at the four corners of a square and a charge q is at its centre. If the system is in equilibrium the value of q is:

1) $-\frac{Q}{4}(1+\sqrt{2})$
2) $\frac{Q}{4}(1+2\sqrt{2})$

3) $-\frac{Q}{2}(1+2\sqrt{2})$

4) $\frac{Q}{2}(1+2\sqrt{2})$

14. Three charges $-q_1+q_2$ and $-q_3$ are placed as shown in the figure. The $x-$component of the force on $-q_1$ is proportional to:



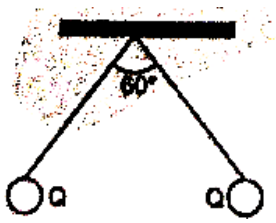
1) $\frac{q_2}{b^2}-\frac{q_3}{a^2}\sin\theta$

2) $\frac{q_2}{b^2}-\frac{q_3}{a^2}\cos\theta$

3) $\frac{q_2}{b^2}+\frac{q_3}{a^2}\sin\theta$

4) $\frac{q_2}{b^2}+\frac{q_3}{a^2}\cos\theta$

15. Two small spherical balls each carrying a charge $Q=10\mu C$ (10 micro-coulomb) are suspended by two insulating threads of equal lengths 1m each, from a point fixed in the ceiling. It is found that in equilibrium threads are separated by an angle 60° between them, as shown in the figure. What is the tension in the threads? (Given $\frac{1}{(4\pi\varepsilon_0)}=9\times10^9\,Nm^2/C^2$).



1) 18 N
2) 1.8 N
3) 0.18 N
4) None of these

16. Three charges each of magnitude q are placed at the corners of an equilateral triangle, the electrostatic force on the charge placed at the centroid is (each side of triangle is L):

1) Zero

2) $\frac{1}{4\pi\varepsilon_0}\cdot\frac{q^2}{L^2}$

3) $\frac{1}{4\pi\varepsilon_0}\cdot\frac{3q^2}{L^2}$

4) $\frac{1}{12\pi\varepsilon_0}\cdot\frac{q^2}{L^2}$

17. A pendulum bob carries a negative charge $-q$. A positive charge $+q$ is held at the point of support. Then, the time period of the bob is.

1) Greater than $2\pi\sqrt{\frac{L}{g}}$

2) Less than $2\pi\sqrt{\frac{L}{g}}$

3) Equal to $2\pi\sqrt{\frac{L}{g}}$

4) Equal to $2\pi\sqrt{\frac{2L}{g}}$

18. A charge q_1 exerts some force on a second charge q_2. If third charge q_3 is brought near, the force that q_1 exerts on q_2 and net force on q_2 respectively

1) Decreases, increases
2) Increases, increases
3) Remains unchanged, may increase or decrease
4) Remains unchanged, remains unchanged

19. Three charges $4q, Q$ and q are in a straight line in the position of $0, l/2$ and l respectively. The resultant force on q will be zero, if Q is:

1) $-q$
2) $-2q$
3) $-\frac{q}{2}$
4) $4q$

20. Two equal charges are separated by a distance d. A third charges placed on a perpendicular bisector at x distance will experience maximum coulomb force when:

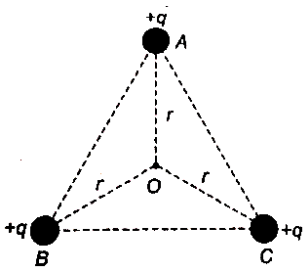
1) $x = \frac{d}{\sqrt{2}}$

2) $x = \frac{d}{2}$

3) $x = \frac{d}{2\sqrt{2}}$

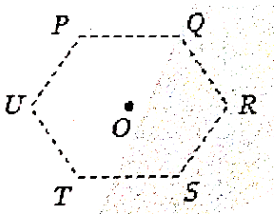
4) $x = \frac{d}{2\sqrt{3}}$

21. ABC is an equilateral triangle. Charge +q are placed at each corner. The electric intensity at O will be:



1) $\frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}$
2) $\frac{1}{4\pi\varepsilon_0}\frac{q}{r}$
3) Zero
4) $\frac{1}{4\pi\varepsilon_0}\frac{3q}{r^2}$

22. Six charges, three positive and three negative of equal magnitude are to be placed at the vertices of a regular hexagon such that the electric field at O is double the electric field when only one positive charge of same magnitude is placed at R. Which of the following arrangements of charges is possible for P, Q, R, S, Tand U respectively?



1) +,−, +,−,−+
2) +, −, +, −, +, −
3) +, +, −, +−, −
4) −, +, +, −, +, −

23. Two small spherical shell A and B are given positive charge of 9C and 4C respectively and placed such that their centres are separated by 10m. If Pis a point in between them where the electric field intensity is zero, then the distance of the point P from the centre of A is.

1) 5 m
2) 6 m
3) 7 m
4) 8 m

24. A particle of mass 20g and charge -3 mC moving with a speed 40 m/s enters a region of electric field 80 N/C in the same direction of the electric field. Speed of the particle after 3 seconds is:

1) 76 m/s
2) 4 m/s
3) 40 m/s
4) 12 m/s

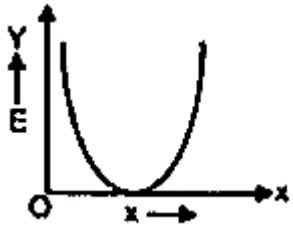
25. An a-particle of mass $6.4\times10^{-27}\,kg$ and charge $3.2\times10^{-19}C$ is situated in a uniform electric field of $1.6\times105Vm^{-1}$. The velocity of the particle at the end of $2\times10^{-2}m$ path when it starts from rest is:

1) $2\sqrt{3}\times10^{5}ms^{-1}$
2) $8\times10^{5}ms^{-1}$
3) $16\times10^{5}ms^{-1}$
4) $4\sqrt{2}\times10^{5}ms^{-1}$

26. A positively charged particle moving along X-axis with a certain velocity enters a uniform electric field direction along positive Y-axis. Its:

1) Vertical velocity changes but horizontal velocity remains constant
2) Horizontal velocity changes but vertical velocity remains constant
3) Both vertical and horizontal velocities changes
4) Neither vertical nor horizontal velocity changes

27. Two identical point charges are placed at a separation of d. Pis a point on the line joining the charges, at a distance x from any one charge. The field at Pis E, E is plotted against x for value of x from close to zero to slightly less than d. Which of the following represents the resulting curve?

1)


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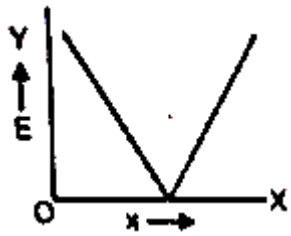

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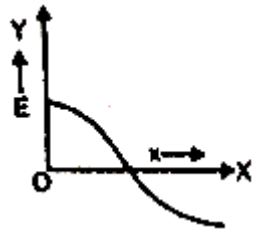

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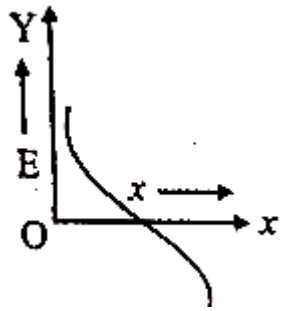

28. An electron of mass M_e initially at rest, move through a certain distance in a uniform electric field in time t_1. A proton of mass M_p also initially at rest, takes time t_2 to move through an equal distance in this uniform electric field, neglecting the effect of gravity, the ratio t_1/t_2 is nearly equal to:

1) 1
2) $\sqrt{M_p/M_e}$
3) $\sqrt{M_e/M_p}$
4) 1836

29. The distance between the two charges 25μC and 36μC is 11 cm. At what point on the line joining the two, the electric field intensity will be zero.

1) At a distance of 5 cm from 25μC
2) At a distance of 5 cm from 36μC

3) At a distance of 10 cm from 25μC
4) At a distance of 11 cm from 36μC

30. A charge of +2μC is placed at a point P where the electric field due to other charges has a magnitude of $3\times10^{6}\ N/C$ and directed to the right. If the charge is replaced with a charge of-2μC then the electric field at point P.

1) decreases in magnitude and changes direction
2) increase in magnitude and changes direction
3) remains the same
4) has the same magnitude but changes direction

Answer Key

(1)	2	**(16)**	1
(2)	4	**(17)**	3
(3)	3	**(18)**	3
(4)	2	**(19)**	1
(5)	4	**(20)**	3
(6)	4	**(21)**	3
(7)	4	**(22)**	4
(8)	2	**(23)**	2
(9)	3	**(24)**	2
(10)	2	**(25)**	4
(11)	2	**(26)**	1
(12)	3	**(27)**	4
(13)	2	**(28)**	3
(14)	3	**(29)**	1
(15)	2	**(30)**	3